BEDSIDE COGNITIVE ASSESSMENT

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Introduction

Bedside cognitive assessment is the assessment of cognition at the bedside without the use of special tools.[1-3] It provides a rough estimate of cognitive ability. Neuropsychological assessment is more sensitive and the gold standard, but often requires preparation and the use of special instruments.

Goals of bedside cognitive assessment:

1. To document the presence and extent of cognitive impairment.
2. To describe the profile of cognitive impairment and aid with the differential diagnosis.
3. To monitor disease progression and response to treatment.
4. To screen for the need of a comprehensive neuropsychological assessment.
5. To provide information regarding cognitive functioning in a forensic context.
6. To aid in the assessment of capacity to make decisions.

Bedside cognitive assessment is especially useful to determine the extent and profile of cognitive impairment. It is best used for mild to moderate impairment. Very mild cognitive impairment should be assessed with neuropsychological tests. Severe cognitive impairment can be assessed by the Mini-Mental Status Examination (MMSE) test.[4]

Measuring the extent of cognitive impairment is important as it describes the degree of change from normal and allows the clinician to include this information in a management plan.

Estimating cognitive ability at the bedside should be done with bedside tests as casual conversation and even routine psychiatric mental status assessment may not reveal true cognitive ability. Many clinicians will be surprised at the extent of cognitive deficit revealed when dedicated bedside cognitive measures are used to estimate a patient's cognitive or intellectual capacity.

The profile of impairment helps us to diagnose the specific condition responsible for causing the cognitive deficit. Significant memory impairment in an older adult person with few other cognitive deficits is suggestive of Alzheimer's disease in the early stages of the illness. Mild memory deficits together with moderate executive dysfunction in the context of cerebrovascular risk factors are more suggestive of vascular dementia. In schizophrenia bedside assessment may reveal impaired attention, executive dysfunction, and mild memory difficulties that include problems with working memory.

Cognitive assessment also helps to monitor disease progression and response to treatment by repeating the assessments to monitor for any changes. An example is the monitoring of disease progression in early to middle stages of Alzheimer's disease with a verbal memory task such as a 10-word list. Recalling all 10 words gives a score of 10 and with three trials a score out of 30 can be obtained. The task can be repeated at 3-6 monthly intervals with different word-sets to limit a learning effect. The Alzheimer's Disease Assessment Scale (ADAS) incorporates this task

and is used extensively to evaluate the effectiveness of cholinesterase inhibitors in the treatment of dementia.[5] See the section on memory for a further description of the task.

Bedside cognitive assessment is not as sensitive as neuropsychological assessment and may serve only as a screening tool for a more comprehensive cognitive assessment. Individuals with higher premorbid intellectual ability presenting with very mild cognitive deficits are best assessed by neuropsychological tests. For moderate to severe deficits, bedside tests have an advantage over neuropsychological assessment because the decreased sensitivity of bedside tests produces more relevant information and is therefore easier to interpret. It creates less noise and helps to focus on actual deficits.

Clinicians often have to consider an individual's cognitive ability when providing an opinion in a forensic context. Bedside cognitive assessment is a useful screening tool and can obviate the need for further assessment. Standardised assessment batteries such as the Montreal Cognitive Assessment (MOCA) are preferred as robust standardised norms are important in a forensic context.[6]

Bedside cognitive assessment provides useful information when evaluating for guardianship and testamentary capacity. Decisions regarding medical treatment, lifestyle and finances are heavily influenced by an individual's cognitive ability. If a bedside assessment is within normal limits then other variables, such as psychosis or personality, needs examining in more detail for determination of capacity. If the bedside assessment indicates moderate to severe cognitive impairment then an individual's capacity is possibly impaired. This should be explored further by

evaluating an individual's thinking for a specific decision. It is possible to have a degree of cognitive impairment, but still be rational with specific decisions such as consenting to a surgical procedure.

Normal vs. impaired cognitive ability

Cognitive impairment is diagnosed only if there is a decrease in cognitive ability from a premorbid level.

Premorbid cognitive ability can be estimated by asking about years of schooling and further education. Occupation and most recent employment provide further information. Knowing the education and occupation of parents and siblings may also help. Based on this information premorbid cognitive function is categorized in one of three categories; below average, average and above average. Most standardised norms have been established for average premorbid intellectual function. Discretion needs to be applied for above average and below average premorbid cognitive ability as the norms for bedside tests may not always be applicable.

Age affects cognitive performance and should be considered when interpreting results in an older adult patient. For instance, the expected MMSE test result decreases significantly with age. The usual cut score for the MMSE is <25, this changes to 23 at about age 80 and is even lower for persons older than 85.[7]

Elements of cognition

The brain is the seat of cognition. The nervous system is designed to interact with the environment through sensory and motor components. Sensory systems provide input to the brain and output is through motor systems.

In-between the input and output systems neurocognitive processing occurs which can be categorised into six basic elements or domains:

1. Attention and concentration
2. Language
3. Executive or planning functions
4. Memory
5. Higher order sensory processing or gnosis
6. Higher order motor processing or praxis

The brain processes sensory input primarily in the primary sensory gyrus and this information is passed on to unimodal and heteromodal association areas. To perform motor acts, the heteromodal association areas project to unimodal motor association areas that in turn project to the primary motor gyrus. From here the muscles are directed to perform actions.

Executive functioning refers to the sequencing and planning of cognitive tasks. It also involves problem solving, decision-making and cognitive flexibility. Executive functioning is a distributed cognitive ability as it modulates many aspects of cognition, including gnosis, praxis, and memory tasks.

Attention and concentration is another distributed cognitive task. Impairment of attention can affect all aspects of cognition.

See Table 1 below for a summary of the 6 elements of cognition with a description and clnical equivalent.

Table 1

Elements of Cognition	Description	Clinical Equivalent
Attention	Selecting and maintaining focus.	Attention and Concentration
Language	Expressive and receptive components.	Aphasia
Planning	Problem solving, decision-making and cognitive flexibility.	Executive Function
Memory	Holding information for online processing and storing for recall.	Amnesia
Sensory systems	Higher order processing of vision, auditory, olfaction, taste, touch, vibration, proprioception, pain, temperature information.	Agnosia
Motor systems	Higher order control of skilled movement.	Apraxia

Measuring the elements of cognition and grading the strength and weakness of each element will provide a profile of cognitive ability and enable to classify the type of dementia. In an older adult patient, impaired memory on bedside assessment with

relative preservation of other cognitive abilities indicates possible early Alzheimer's disease. Whereas, early language impairment with no other cognitive problems may indicate primary progressive aphasia. Poor executive functioning with behavioural changes may indicate a behavioural variant of a fronto-temporal dementia.

The fifth edition of the Diagnostic and Statistical Manual of Mental Disorders, (DSM-5) defines criteria for dementia (also referred to as major neurocognitive disorder), mild neurocognitive disorder and delirium.[8] The criteria include the elements of cognition as described above. Gnosis and praxis is combined and described as perceptual-motor function. DSM-5 also includes a neurocognitive element called social cognition, which it defines as the recognition of emotions and a theory of mind.

Cognitive tasks that make up a cognitive test

Most of the individual tasks that comprise a bedside cognitive assessment battery measure more than a single element of cognition. For instance, orientation to time on the MMSE is a memory test, but it also involves language and attention. Language is involved because the patient has to first understand the question before responding in an understandable language. Attention and concentration is involved to focus on learning the date and time. A patient with delirium may not concentrate when informed about the date and hence will not retain and recall this information. They also provide incorrect answers if attention is impaired during the assessment as they may not have registered the questions asked.

Examine for themes or patterns of impairment after the completion of bedside cognitive assessment. The total MMSE and MOCA score will only indicate general cognitive performance. If the test items are examined in more detail then a pattern of impairment becomes apparent which will help with the differential diagnosis. Hypotheses about a diagnosis can be further examined by using some of the bedside tests described below.

An example is a 70-year-old university graduate with a 6-month history of progressive cognitive impairment. The patient scores 28/30 on the MMSE losing 2 points on delayed recall in the 3-word recall test. An above average premorbid IQ would be expected in this person and not being able to recall 2 of the 3 words indicates impairment. A verbal memory deficit with a history of progressive cognitive decline indicates possible early Alzheimer's disease. This hypothesis can be further examined with more sensitive verbal memory tests such as a 10-word recall test as described in the section on memory below.

Procedural aspects of cognitive assessment

A number of other factors influence the assessment and interpretation of bedside tests.

Correct technique is essential when performing cognitive assessment for the test results to be reliable. Sufficient time with no interruptions is also essential during the assessment.

Bedside assessment should not include tests that are used in neuropsychological testing as it can induce a practice effect and thus complicate the interpretation of neuropsychological tests for a neuropsychologist.

See below for a description of the individual cognitive tasks.

Attention & Concentration

Introduction

Attention is the ability to focus on a stimulus and concentration is the ability to maintain that focus.

Impairment of attention and concentration can be measured with tests of different levels of sensitivity. It can vary from simple observation of the patient to specialised computer based assessments.

A patient's response to stimuli can be observed and described. For example, “The patient opened her eyes when her name was called aloud.” Or, "The patient appeared sleepy and could not be aroused when her shoulder was touched." Consciousness can then be further described on a continuum of; awake, drowsy, stuporous, and comatose.

The Glasgow Coma Scale (GCS) is used for a more accurate description when a patient is stuporous or comatose. If the GCS is 15/15 but cognitive impairment is still suspected then further bedside assessment with tests for attention is necessary.

Applying a continuous stimulus and measuring the ability to sustain focus can measure attention and concentration. The patient has to hold the information active and process it continuously. This is also called online processing which overlaps with working memory. Whilst maintaining focus, the patient has to ignore distractions and avoid switching tasks, which also overlaps with executive functioning. Thus, there are close functional and anatomical relationships in the brain between the following neurocognitive functions:

- Attention and concentration
- Working memory
- Executive functioning

Apparent poor performance on tests for attention and concentration may be related to impairment of working memory or executive functioning. Further assessment of working memory and executive functioning will shed light on the most important deficit.

Attention and concentration can fluctuate over time and this variability must be examined for by consulting with the patient more than once. This can be done later the same day or at a different time the next day.

Localisation

Normal attention and concentration requires optimal interaction between the reticular formation, thalamus and neocortex.

Tests

The vigilance test described below is the best assessment for attention and concentration, as unlike the other tasks, it does not significantly overlap with working memory and executive functioning.

1. Observation

It is possible to assess attention and concentration by observing the patient's quality of interaction during mental status examination. However, this provides only a rough estimate and is a precursor for further testing.

2. Spelling of a 5 letter word backwards

First ask the patient to spell the word forwards to verify correct spelling. Then ask the patient to spell the word backwards. Some experts advocate the use of silent letters in a word, such as in "world" and "right." However, asking the patient to spell the word forwards first may negate the difficulty of a silent letter. In many situations, words like "April" or "March" is a good choice as these words are familiar.

Most individuals can spell a five-letter word backwards without any mistakes.

3. Serial 7's or 3's

For serial 7's the patient is asked to: "Subtract seven from one hundred." Wait for the response. "Now subtract seven again and continue to do this until I ask you to stop." For serial 3's start with 20.

Most individuals can do the serial 7's and 3's correctly.

Video illustration: https://vimeo.com/162259525

4. Vigilance test

Inform the patient: "I am going to say a long series of letters. Whenever you hear the letter A, tap the desk. If you hear any other letters then do not tap." Say a random series of letters at a rate of about 1 letter per second with the letter A occurring regularly as in the following example: "K T B A O S A W I E A A G H C A T A A A E ..." Monitor the number of A's that were correctly indicated by a patient tapping the desk and note the number of additional taps. The additional taps may also indicate an inability to inhibit an unwanted response which is called perseveration.

Most people complete this task without any errors.

Video illustration: https://vimeo.com/162247583

5. Digit span

This can be done forwards and/or backwards. The example given here is for forwards.

"I will now say a series of numbers and would like you to repeat them. Are you ready? 4 9." Note the response and continue by increasing the series with an extra number until a sequence of 9 is reached. Most individuals can repeat a sequence of 6-8 random digits.

If a patient fails the sequence then repeat it with another random sequence of the same length. This helps to confirm the cognitive limit for the patient and also excludes other reasons for failing, such as not having heard the instruction well or misunderstanding the instruction.

Video illustration: https://vimeo.com/158806267

Language

Introduction

Language is the production of sounds that form understandable words, sentences and ultimately discourse. It also refers to the ability to comprehend, to read and to communicate in writing.

Localisation

In more than 98% of right-handers the perisylvian language areas of the left hemisphere are responsible for language processing. This includes Wernicke's area, the arcuate fasciculus, Broca's area and other associated cortical areas. The right hemisphere is involved in other aspects of language such as humour, metaphor and determining the emotional content of language.

A comprehensive assessment of language involves assessment of the following six aspects of language:

1. Spontaneous speech
2. Comprehension
3. Repeating words or phrases
4. Naming objects
5. Reading
6. Writing

Tests

1. Spontaneous speech

To assess spontaneous speech the patient is allowed to speak freely and the quality of speech is assessed. This can be done during the standard psychiatric assessment. However, it is often difficult to get aphasic patients to speak freely as their ability to produce language is often impaired. In these situations it is easier to elicit spontaneous speech by showing the patient a complex picture and asking the patient to describe what is happening in the picture. The well-known “Cookie Jar Theft” picture is an example.

Abnormality in spontaneous speech is indicated by impaired fluency, articulation or the presence of paraphasias. Paraphasias are word substitutions and can be phonemic i.e. “sitter” for “sister” or semantic i.e. “jug” for “glass.” Word finding difficulties are associated with stock phrases such as “thingy” or “the whatsit” to replace intended words.

2. Comprehension

2.1 Stage 1-4 Commands

Provide the following verbal commands without any non-verbal clues such as hand gesturing. “Point to the ceiling.” Assess the response and continue. “Point to the ceiling and then to the floor.” Again, assess the response and continue. “Point to the ceiling, the floor and then to the door.” Again, assess the response and continue. “Point to the ceiling, the floor, the door and then to the window.” Most patients can complete all 4 stages.

A simpler way to screen for impairment is to give a 3-stage command: “Point to the ceiling, floor and the door.” If the response is compromised then follow the above-mentioned method for a more detailed assessment.

Video illustration: https://vimeo.com/158806262

2.2 Moving 1-3 Objects

Place 3 objects in front of the patient and name them while they are being placed to exclude naming ability and object agnosia as confounding factors. Common objects such as a coin, keys and a piece of paper can be used. Begin with simple commands and then move to more complex instructions. “Turn the coin over”, “Place the keys on the other side of the paper”’ and “Place the coin between the keys and the paper, and turn the paper over.”

Video illustration: https://vimeo.com/158806264

3. Repetition

Ask the patient to repeat words, phrases or a sentence. The MMSE test has a phrase for repetition: “No ifs ands or buts.” Other sentences can also be used such as: “The little boy went to the shopping mall to spend his money.” The MOCA test provides the following examples: “I only know that John is the one to help today”, and “The cat always hid under the couch when dogs were in the room.”

4. Naming

Point towards objects and ask the patient to name them. Most people identify the following: watch, pen, tissue, button, shoe, and shirt. Less common objects would be: knuckles, ring finger, eyebrow, winder of a watch, and collar. See below (in the

section on agnosia) for information on how naming ability overlaps with visual object agnosia.

5. Reading

Ask the patient to read words and sentences. The MMSE provides the sentence "Close your eyes." The MMSE requires the patient to read, understand and then respond to the task before a point is scored. If the patient struggles with this task then further assessment of comprehension is indicated.

Addenbrooke's Cognitive Examination (ACE) provides a task for reading ability.[9] The patient is asked to read the following words: sew, pint, soot, dough and height.

6. Writing

Writing ability is tested by asking the patient to write words or sentences. The MMSE provides a task to assess this ability. The patient is asked to write a sentence of their choice. An understandable sentence with subject, verb and object is required for a score.

Aphasia

The most common type of aphasia is a mixed aphasia. Mixed aphasia is a language disorder wherein there is an impaired ability in two or more of the language modalities. Impairment in a specific language modality has various descriptions as indicated in Table 2 below.

Table 2

Language modality	Syndrome associated with impairment
Spontaneous speech	Broca's aphasia, motor aphasia, nonfluent aphasia, expressive aphasia
Comprehension	Wernicke's aphasia, sensory aphasia, fluent aphasia, receptive aphasia
Repetition	Conduction aphasia
Naming	Nominal aphasia
Reading	Alexia
Writing	Agraphia

Executive functions

Introduction

Executive functions are a complex set of cognitive abilities that are involved in planning and multitasking. These functions operate in non-routine situations and exert a top-down, volitional control over cognition and behaviour. Executive functions also refer to the ability to monitor environmental input and to change cognition or behaviour if required. Perseveration is the result of an inability to change. Generating thoughts and actions are also implicated in executive functioning and their impairment manifests as a lack of initiation or apathy.

Localisation

The frontal cortices are involved in executive functioning and it connects closely with subcortical structures. These connections form loops projecting from the frontal cortex to the striatum, globus pallidus, thalamus and back to the frontal cortex. A lesion in any part of this circuit will result in deficits that are indistinguishable from a frontal cortical injury. The anterior cingulate has many connections with the frontal lobes and is also involved in executive functioning.

Tests

1. Luria Hand Movements I

A Russian neuropsychologist, AR Luria, first described alternating hand movements as a measure of executive functioning.[10]

For Luria I refer to Figure 1.

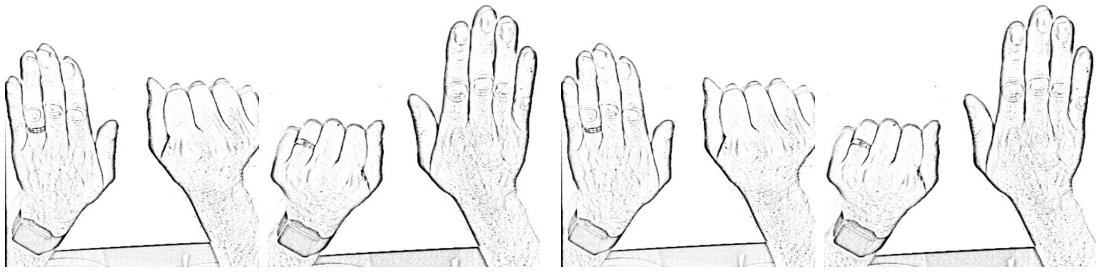


Figure 1

The instruction to the patient is to: "Place both hands on the table like I do. Watch me carefully because I want you to do the same as I am doing. Can you see that I am making a fist with one hand? Now I swap it around. Can you do that as fast as you can?" Execute five cycles of alternating movements with the patient following the examiner. Then let the patient continue with the task and count the number of cycles completed without an error. Most individuals can complete 5 cycles.

Video illustration: https://vimeo.com/161349540

2. Luria Hand Movements II

The examiner performs alternating movements with the right hand by making a fist, then opening the hand with the edge touching the surface, and then the palm touching the surface. The series fist-edge-palm is repeated. See Figure 2 below.

The instructions are: "Can you see what I am doing? Please join me and do the same." The patient should be able to perform 5 cycles correctly. Remember to test

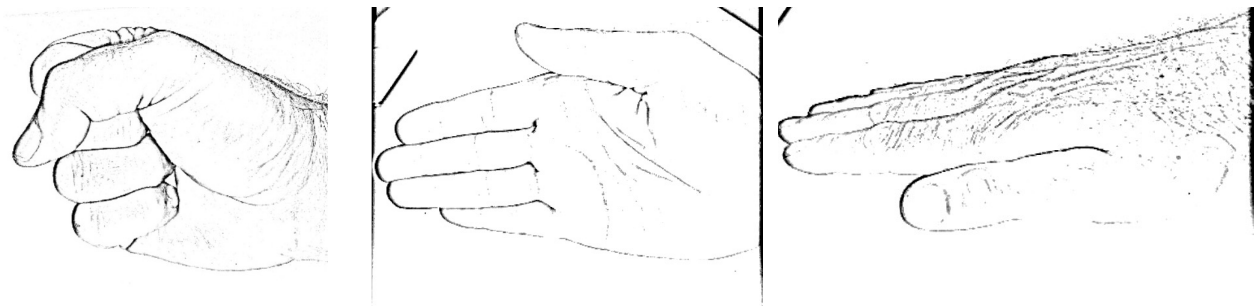


Figure 2

both hands.

Video illustration: https://vimeo.com/161349541

3. Word Fluency Test

Word fluency tests measure problem solving. The patient has to consider how to generate words in a certain category. For instance, with the animal naming test the patient may start naming pets, then think of farm animals, then wild animals on the African plains, and etc.

Word fluency tests are sometimes included as part of language assessment as poor fluency is associated with motor aphasia. Impairment on fluency tests also correlates with lack of initiation of behaviour and apathy.

3.1 Animal Naming Test

“I will give you one minute to name as many animals as possible. Any animal will do. Are you ready? Go." The patient can name mammals, birds, reptiles or fish. Repetitions are not scored. Note how many words are generated in the first and last 30 seconds. Individuals with brain injury often generate a few words quickly in the first 10-20 seconds and then struggle to generate more words in the last 30 seconds. Most individuals generate more than 12-14 animals per minute.

Video illustration: https://vimeo.com/158806167

3.2 Lexical Fluency Test

"Say as many words as you can beginning with the letter 'S', any words except names of people or places. I will give you one minute to complete the task. Are you ready? Go." Repetitions are not scored, but should be noted as it may indicate

perseveration. Most individuals generate 12 or more words. Again, note how many words are given in the first and last 30 seconds. The letters 'F' and 'A' can also be used.

4. Design Fluency Test

The design fluency test is a visual equivalent of word fluency tests and measures the ability to generate designs.[11] This test is probably more sensitive to non-dominant frontal lobe pathology. Word fluency tests are more closely associated with dominant frontal lobe pathology.

“You have one minute to draw anything using four lines. The drawings must be different and I must be able to count the lines. I will give you two examples.” Draw a hash sign and steps as indicated in Figure 3 below. Count the lines as you draw them. “Do you have any questions? Your minute starts now." Most people can

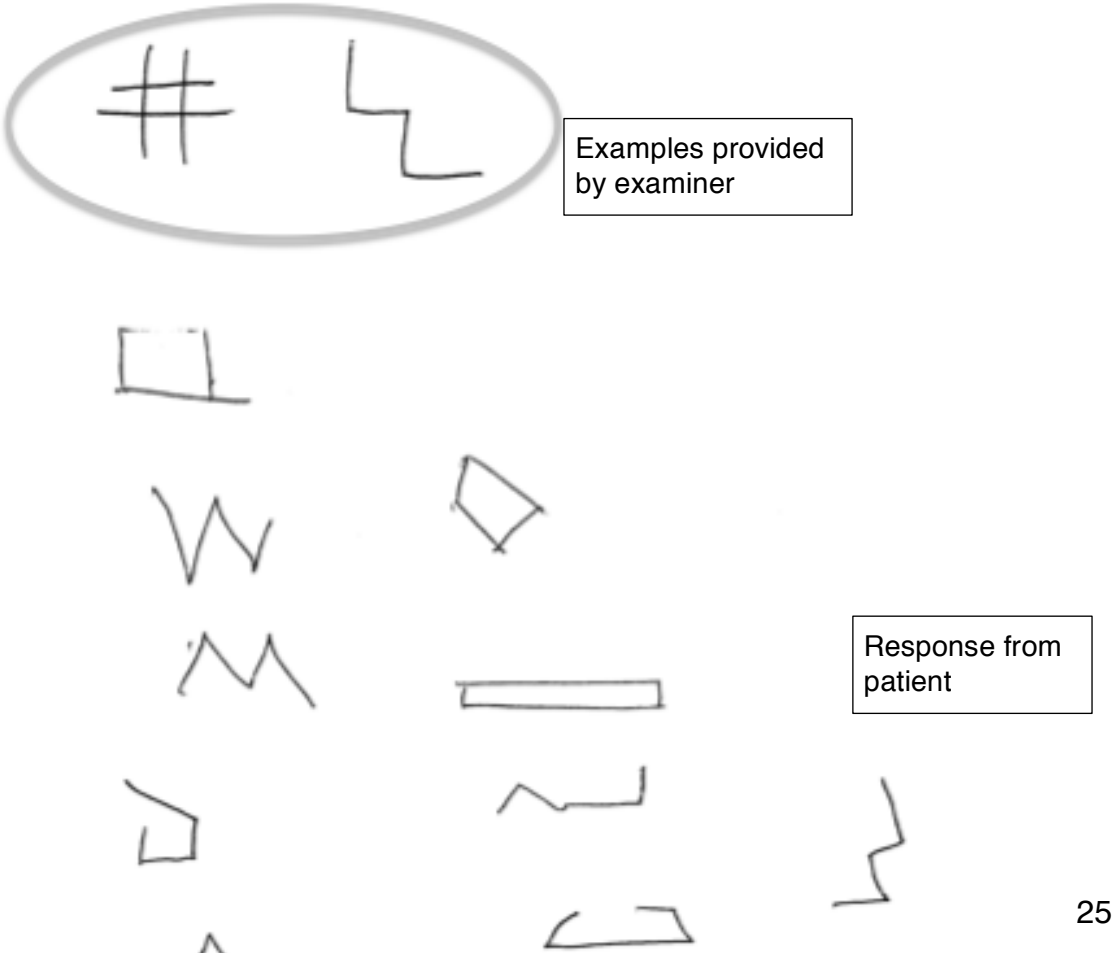


Figure 3

generate 5-7 different drawings, depending on their educational level. See Figure 3 for an example of the test.

Video illustration: https://vimeo.com/158806260

5. Go-No-Go Tasks

The Go-No-Go tasks measure the ability to inhibit or switch behaviour. Perseveration is induced on these tests.

5.1 Finger-Nose Task

The examiner lifts their finger and in response the patient touches their nose. When the examiner touches his nose the patient responds by lifting his finger. The instructions are: "Listen carefully. When I touch my nose, you raise your finger like this. When I raise my finger you touch your nose. Let's try it now." The examiner performs five variations such as *Finger-Nose-Nose-Finger-Nose.*

Most individuals can complete this task without making a mistake.

Video illustration: https://vimeo.com/158806269

5.2 Tap Task

"Tap once when I tap once. Do not tap when I tap twice." Perform a random series of tapping once or twice and note the patient's response. Most people can do this without any mistakes.

Video illustration: https://vimeo.com/162241521

6. Alternating Sequences

Alternating sequences measures the ability to switch thinking and shows cognitive flexibility. It also involves working memory and attention as concepts are held in memory and then manipulated.

"I will say a sequence of letters and numbers. Listen carefully as I would like you to complete it for me. A 1 B 2 C...? Good, now start from A and perform this sequence until I ask you to stop." Score the following sequence: A 1 B 2 C 3 D 4 E 5. Most people can complete this task without any mistakes.

Video illustration: https://vimeo.com/158805839

7. Months of the Year Backwards

The months of the year task is similar to the alternating sequences task as it involves cognitive flexibility and working memory.

"Say the months of the year starting with January." Allow the person to complete the task until December. "Now start with January and go backwards." Stop the test when August is reached or 2 or more mistakes are made. Most people can complete the task without any errors.

Memory

Introduction

Memory refers to the ability of the brain to store information for their use at a later time. The process involves encoding, storage and retrieval of information.

Localisation

The hippocampus and parahippocampal areas situated in the medial temporal lobes are primarily involved in the storage of information, while the frontal cortex is involved in retrieval. The basal ganglia, cerebellum and cortex are also involved in memory functions.

Working memory is known as our 'online' memory and refers to the ability to store and manipulate information for up to 30 seconds. It involves a network of frontal cortical and subcortical circuits that communicate with a central executive. It has a verbal component in the dominant hemisphere and a visual component in the non-dominant hemisphere.

Short-term and long-term memory that is explicitly processed through conscious effort is called declarative memory and can be further subdivided into episodic and semantic memory. Episodic memory is the conscious recollection of experiences framed in our own context. An example is remembering what you did last night. Semantic memory is our general store of conceptual and factual knowledge. Remembering the date of the Second World War or the chemical formula for water are examples. The initial memory trace for semantic memory is episodic in nature, but the contextual information is lost over time. Thus, we remember the chemical formula for water but forget where or how we learned it.

Implicit or non-declarative memory is recalled without conscious reflection and involves procedural memory. This is the type of memory that is used for remembering how to ride a bicycle.

Timing the delay to recall information presents another way of classifying memory. This is done by providing a stimulus, either verbal (written or spoken format) or visual, and then testing for recall after a period of time. For working memory, allow up to a minute before asking for recall. For short-term memory, allow minutes to hours. Long-term memory is usually not assessed by the bedside. To measure it you need to allow weeks or months to pass before recall is examined.

The classification of memory is not consistent across textbooks. Episodic memory is sometimes used to refer to short-term memory. Episodic memory is also sometimes subdivided into verbal and visual episodic memory.

See Table 3 and 4 below for examples of how memory is classified.

Table 3

Classification of Memory According to Time Elapsed Since Stimulus Presentation	
Working Memory	Verbal Visual
Short-term Memory	Verbal Visual
Long-term Memory	

Table 4

Classification of Memory According to the Process of Recall		
Explicit Memory	Working Memory	Verbal Visual
Implicit Memory	Episodic	Verbal Visual
	Semantic	
	Procedural memory, priming, conditioning	

Tests

Working memory

1. Verbal Working Memory

Verbal working memory is assessed by providing a verbal stimulus and immediately asking for recall. The stimulus can be words or numbers. Working memory overlaps considerably with attention and concentration. Digit Span is a test for working memory based on numbers and is described above in the section on assessment of attention and concentration.

Immediate recall of 3 words

“I would like you to remember 3 words, listen carefully as I will say the words only once. Rock, house and flag. Can you tell me what those three words were?” Most individuals can immediately recall all three words. In most instances this test progresses to a delayed recall task. If the patient cannot immediately recall all three words then it is permissible to repeat the words 4-6 times. If the patient is still unable to repeat the words then do the next test. Warn the patient about the delayed recall task: “Remember these three words as I will ask for them later on.” It is advisable to use three unrelated and unfamiliar words. Avoid using the same words used in the MMSE as patients may have heard it previously and a practice effect will make the task easier.

Immediate recall of 10 words

“I am going to show you a list of words to read and remember. I will show you the same list twice and ask what you can remember each time. Later on I will ask you if you can still remember the words. Are you ready? I will page for you.” Allow the patient to read the words from cards or a computer monitor. The list of words is provided below. “Which of the words can you remember?” Most people can remember 5 or more words on the first trial.

Another trial of the same words but in a different order can be done. This is strictly speaking not a verbal working memory task as most people begin to rely on memory strategies and less ‘online’ abilities for recall. It starts to overlap with short-term memory.

“Thank you. Please read the words again while you try to remember them. What can you remember now?” Most people will show a learning curve and are able to build on the first recall task to remember 6 words or more.

List of Words for Verbal Recall:

Trial I

CARROT BOAT LION EYE JUG CAT SOUP LEG HOUSE TICKET

Trial II

LION JUG EYE BOAT CARROT LEG TICKET HOUSE CAT SOUP

2. Visual Working Memory

This can be assessed by showing the patient a line drawing for 10 seconds, removing the drawing and waiting for another 10 seconds, and then asking the patient to draw what was seen. See the Figure 4 below from Strub and Black for 3 examples. [12]

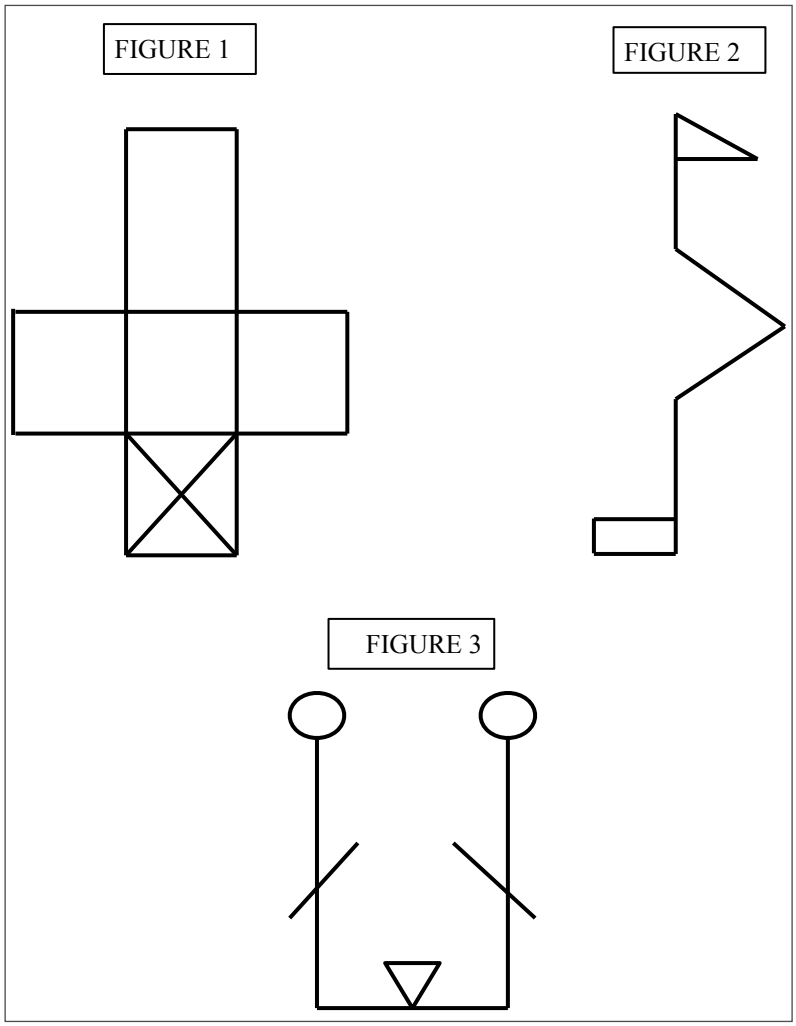


Figure 4

The response of the patient can be scored with a simple scoring system ranging from 0-3; 0 being a failure to recall anything, 1 - poor recall, 2 - few omissions, and 3 - no mistakes.

Impaired constructional ability due to apraxia or agnosia will make it difficult to use visual memory tests based on line drawings. This is because the patient is unable to copy the line drawings. Therefore it is important to ensure that constructional ability is normal before interpreting visual memory tests based on line drawings.

Short-term memory

Short-term memory can be assessed by delaying the recall task for a few minutes, ideally for around 5-10 minutes. This is usually done by doing other cognitive assessment tasks in between assessing working memory and short-term memory. Short-term memory can be subdivided into verbal and visual short-term memory.

1. Verbal short-term memory

3 Words delayed recall

"Can you please tell me the three words that we learned earlier?" Most people can recall all three words.

10 Words delayed recall

"Earlier, I showed you some words printed on the cards (or the computer screen). Can you tell me which of those words you remember?" It helps to point to the actual cards or screen. Recollection of 4 or more words is considered normal.

CARROT BOAT LION EYE JUG CAT SOUP LEG HOUSE TICKET

Cued recall

Cueing recall helps patients with frontal lobe injury to retrieve information from their memory banks in the temporal lobes. Better performance on cued recall may indicate a problem with the frontal cortex. If cueing does not improve performance then it indicates a hippocampal problem.

Cued recall can be assessed with a recognition recall task:

“I am now going to read a list of words. Some of the words are from the original list that I showed you, (pointing to the list of words on the examination table or the computer screen helps to remind the patient of the previous list) and some of the words are new and are not from the original list. Please tell me which of the words were from the original list. If you don’t know the answer, just guess." Read the words below. One mark is given for correctly identifying a word from the original list (in bold) and for correctly saying that the word was not in the list (words not in bold). The instructions may be repeated if it is forgotten.

Recognition Memory List

CARROT ONION BUG **JUG** SHIP **BOAT LION** GOAT SPY **EYE SOUP** BUTTER BAT ARM **CAT LEG** MOUSE **TICKET HOUSE** BUCKET

Most people make only one or two mistakes, as cued recall is surprisingly good. However, individuals with medial temporal lobe pathology continue to do poorly despite cueing, whereas patients with frontal lobe memory impairment perform better.

2. Visual short-term memory

Delayed visual recall can be assessed by asking the patient to recall the figures from the visual working memory task described above.

An alternative is to present the constructional ability task, described below under praxis, in the beginning of the assessment. Delayed visual recall can be assessed by asking for recall of the drawings later in the assessment.

Semantic memory

Semantic memory can be measured by asking for historical facts. As this is dependent on educational level, questions for well-known historical facts are asked.

Historical facts test

The Addenbrooke's Cognitive Examination provides examples of a semantic memory test. Most people will be able to give a correct answer for 3 or more of the following questions.

"What is the name of the current Prime Minister?"

"What is the name of the current USA president?"

"Name the USA president who was assassinated in the 1960s?"

"When was the second World War?"

Name and address test

This is a good alternative test for verbal memory as it is short and incorporates both verbal working memory and verbal short-term memory.

"I would like you to remember a name and address, listen carefully as I will say it only once and will ask for the name and address in a short while: Peter Black, 32 Long Street, Albany. Can you repeat that?" Score 1 mark for each of the five components. If the patient has errors on repetition then repeat the phrase up to three times.

Allow 5 minutes to pass while distracting the patient. A good distraction technique is to do the months of the year task described in the section on executive functions.

"Tell me the name and address that you learned earlier." Score 1 mark for each component correctly recalled. The test can be further expanded by giving cues. See the Addenbrooke's Cognitive Examination for an example of this task.

Praxis

Introduction

Praxis is the ability to translate an idea into an action and to monitor the successful completion of an action. In apraxia there is an inability to perform purposeful movements despite the presence of normal elementary motor function.

The terminology used to describe the different forms of apraxia is confusing and not consistently applied across different textbooks. Limb-kinetic apraxia can be seen as an impairment of finger and hand motor function, while ideomotor apraxia is an impairment of hand and limb motor function. Ideational apraxia is an impairment of a learned sequence of motor acts.

Localisation

Planning of movement is initially done with words in the posterior aspect of the dominant hemisphere. These plans are then shunted to the frontal cortex, supplementary motor area and motor cortex. Final execution of movement is done from the primary motor cortex.

Tests

1. Limb-kinetic apraxia

This is an inability to make precise movements with the hand and fingers contralateral to a brain injury. It is assessed by asking the patient to rotate a coin between thumb, index finger and middle finger. Remember to assess both hands. Most individuals are able to complete this task without any errors.

If a coin is not readily available then use a pen and ask the patient to rotate a pen using their fingers. Again, most individuals should be able to complete this without any errors.

Video illustration: https://vimeo.com/158806270

2. Ideomotor apraxia

This is a failure to correctly position and move a limb in space. It can be assessed by asking the patient to mime tasks. In this example the subject is requested to mime making tea, first with his right hand and then with the left hand. “Show with your right hand how you would pour a cup of tea, add sugar and stir it. Now do the same with your left hand."

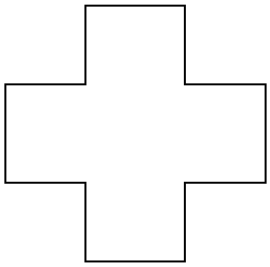
Video illustration: https://vimeo.com/162259526

3. Oral apraxia

Oral apraxia is an inability to perform orobuccal movements. Ask the patient to mime blowing out a match, licking crumbs from the lips and to blow a kiss.

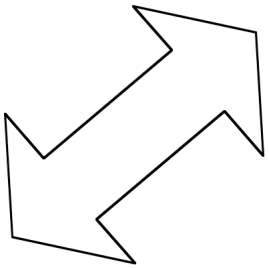
4. Ideational apraxia

This is the ability to correctly sequence a series of well-learned motor acts that lead to a goal. The patient is given an A4 sheet of paper, an envelope and a pencil. “Pretend that you have written a letter and would like to post it. Fold the sheet of paper so that it fits into the envelope, then seal the envelope and address it to yourself and then indicate where you will put a stamp.” The instructions can be repeated, as this is not a test of memory. Most individuals are able to do this task without any difficulty.



Video illustration: https://vimeo.com/158806263

5. Dressing apraxia



This task involves limb-kinetic, ideomotor, and to some extent ideational apraxia. It is assessed by asking the patient to put on a hospital gown or other suitable piece of clothing. Observe if the patient is able to put their hands and arms through the sleeves correctly and is able to fasten the buttons.

6. Constructional ability

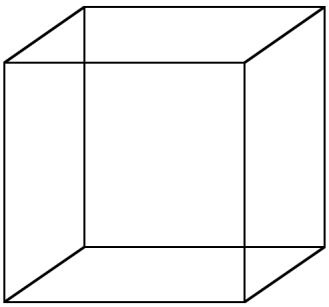


Figure 5

Line drawings are presented and the patient is asked to copy it. The line drawings can be presented in a printed format or sketched by an examiner. Examples of line drawings are a cross, arrows and 3D cube (shown in Figure 5). If the examiner draws the

pictures then take care not to let the patient see how it was drawn as it makes the task easier to complete.

The ability to successfully copy a line drawing depends on correct recognition of the drawing through visual input and processing and then motor function. The input function is part of gnosis. Output is a test of praxis as it involves motor planning and detailed instructions to the dominant hand that draws the picture.

Impaired constructional ability should be further examined to distinguish agnosia from apraxia. The patient can be asked if their drawing resembles the stimulus drawing. If the patient says the drawing differs from the stimulus, but does not know how to draw it correctly then it may indicate apraxia. In this situation the patient is able to perceive a difference between the drawings, but is not able to draw them correctly.

Gnosis

Introduction

Agnosia is an impairment of perceptual functioning due to a break down in higher order processing of sensory stimuli.

Localisation

Higher order sensory function is processed in the sensory association cortices. This involves the parietal lobes and sensory association areas of the occipital and temporal lobes.

Tests

1. Identify objects visually

Visual object agnosia is an inability to identify objects by sight. Ask the patient to identify common and uncommon objects. Most people would be able to identify the following: watch, pen, tissue, button, shoe, and shirt. Less common objects would be: knuckles, ring finger, eyebrow, winder of a watch and collar.

This test overlaps with naming ability and hence further exploration would be required to distinguish between nominal aphasia and visual object agnosia. If the patient struggles to name an object then ask the patient to describe it or to mime its use. If this can be done adequately then perception is usually normal and the deficit is more likely anomia.

Language deficits can also be examined by providing the patient with language cues by saying either a part of the word or giving a phonetic clue. For example, say; “col...” for collar or say it sounds like "dollar." Some patients may then be able to say the word and the deficit is more likely to be a nominal aphasia.

Video illustration: https://vimeo.com/161349545

2. Identify objects by touch

“Please hold out your right hand. I am going to ask you to close your eyes and then I will place an object in your hand. Identify this by touch only.” Place common objects such as keys or a coin in the patient’s hand. Repeat the same with the left hand.

An inability to identify objects by touch is known as astereognosis.

Video illustration: https://vimeo.com/161349544

3. Finger perception test

Ask the patient to place their hands on the table in front of you. “I am now going to touch two of your fingers and you must say how many of your other fingers are in between my fingers.” Touch the fingers indicated in the first picture below, on the black markings. The patient should answer “one.”

“Now close your eyes while I continue this test. How many of your fingers are in between the fingers that I touch?” Continue to touch the patient’s fingers as indicated in Figure 6. Most people would be able to complete this test without any mistakes.

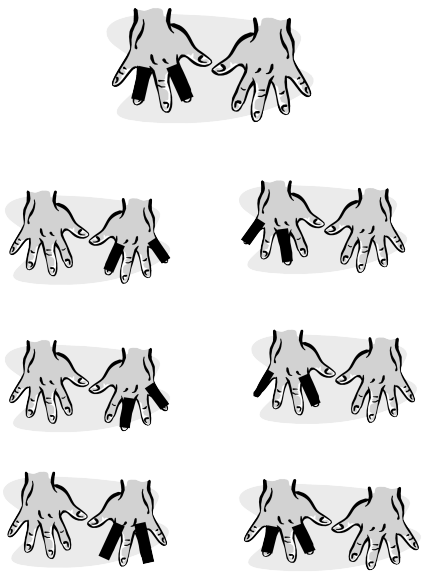


Figure 6

Video illustration: https://vimeo.com/158806265

4. Clock drawing

The clock drawing test is a good measure of general cognitive impairment, but it is fairly non-specific as it does not demonstrate which cognitive domain is impaired.

The different elements of cognition involved in the clock-drawing task include the following. Ability to comprehend, as the instructions are verbal. Drawing is an output function and involves praxis and executive functioning. While the drawing is being done the result is being monitored which in turn guides the completion of the clock drawing. The monitoring process involves visuoperceptual abilities.

Instructions for the clock drawing task are: “Draw a clock face with the time set to 10 past five. It must have all the numbers and hands so that a child can read the time” The instructions may be repeated once if required.

A total of five points are awarded. Most people would be able to score full marks.

1. Score one point if the outer circle is present.
2. Score one point if all the numbers are present, in the correct order and in the appropriate quadrant. No other numbers or letters are allowed.
3. Score one point if all the hands are present. The minute hand must be longer than the hour hand.
4. Score one point if the time is correctly indicated.
5. Score one point for good overall planning (starts with circle and plans the writing of the numbers in appropriate quadrants).

Video illustration: https://vimeo.com/158806261

Suggested Cognitive Assessment Protocols

Bedside tests can be done as stand-alone tasks. However, it is far better to combine tasks into an assessment protocol that measure all the major elements of cognitive functioning. This improves the accuracy and reliability of bedside cognitive assessment.

The MMSE and MOCA tests are good examples of combining different tasks into a cognitive assessment tool or test. The downside of combining tests is that clinicians may focus on the total score of a test and ignore how the elements or domains of cognition are affected. Compare this to a standard clinical neurological examination. A neurologist would not count on a single summary “score” for a neurological examination, instead the findings are interpreted and analysed for a central theme that support a hypothesis for a specific neurological condition.

To avoid missing any impairment in a single domain it is necessary to incorporate tasks that measure all of the cognitive elements in a comprehensive cognitive assessment tool. Localised brain lesions such as stroke or circumscribed tumours may present with impairment in a single cognitive domain.

A cognitive test battery must contain tests that measure cognition at the same level of difficulty. It is not helpful to combine sensitive and rough measures in the same protocol.

Tests must be administered in a specified order. It is usual to start with memory tests so that delayed recall can be done later during the assessment period. Visual memory tasks should not be contaminated by other visual test items. To avoid this, line drawings for other tasks such as constructional ability is administered only after visual memory tests are completed.

The MMSE and MOCA are good tests to start with. These tests help to establish a baseline score, which can be readily shared between clinicians. Tasks can be added as appropriate to measure elements of cognition not adequately addressed in the tests or for further exploration of a specific domain.

Tasks can be added to the MMSE to measure executive functioning, as the MMSE does not thoroughly evaluate executive functioning. The only part of the MMSE that measures executive functioning to an extent is the serial 7s and the test to spell backwards.

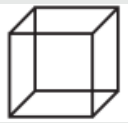
The MMSE can also be complimented with additional tasks for praxis and gnosis. This is particularly relevant when a patient fails the pentagon copy task and to clarify whether the deficit is apraxia or agnosia, or a combination of these.

A suggested short protocol of tests to compliment the MMSE:

- Luria hand movements
- Animal naming task
- Finger-nose task
- Limb-kinetic praxis (coin turning task)
- Identifying and naming common objects

Some patients with a higher educational level may find the easier questions of the MMSE belittling. For such individuals a short protocol without the MMSE may be more appropriate. The protocol as indicated below in Table 5 can be followed.

Table 5

10 Minute Cognitive Assessment without MMSE	
Task	**Result**
Name and Address Memory Test. Peter - Thomson - 32 - Long Street - Albany	
Copy Cube 	
Luria Hand Movements I	
Luria Hand Movements II	
Animal Naming Test	
Design Fluency Test	
Limb-Kinetic Praxis: Coin Task	
Identifying And Naming Objects: knuckles, ring finger, eyebrow, winder of a watch and collar.	
Delayed Recall: Name and Address Memory Test. Peter - Thomson - 32 - Long Street - Albany Recall can also be assisted with cueing.	

The MOCA test does not have a specific task to examine for the ability to comprehend language. It is thus possible to complete the MOCA and obtain a low score without appreciating the contribution of a receptive aphasia. Therefore, it is useful to do a short comprehension task before administering the MOCA and a good option is to do a 3-4 stage verbal comprehension command.

Additional tasks for testing constructional ability and a delayed recall task for visual memory can be added to the MOCA. MOCA can also be expanded (with examples described elsewhere in this text or by using similar tasks from the Addenbrooke's test) for further assessment of nominal aphasia and visual object agnosia. Finally, additional tests for executive functioning and praxis can also be added to the MOCA.

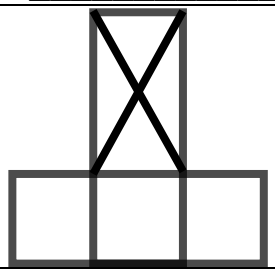
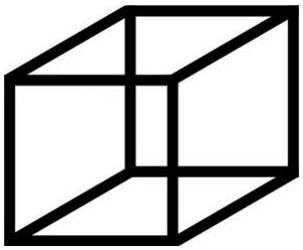
Video illustration for the MOCA: https://youtu.be/ryf8SG0NQLQ

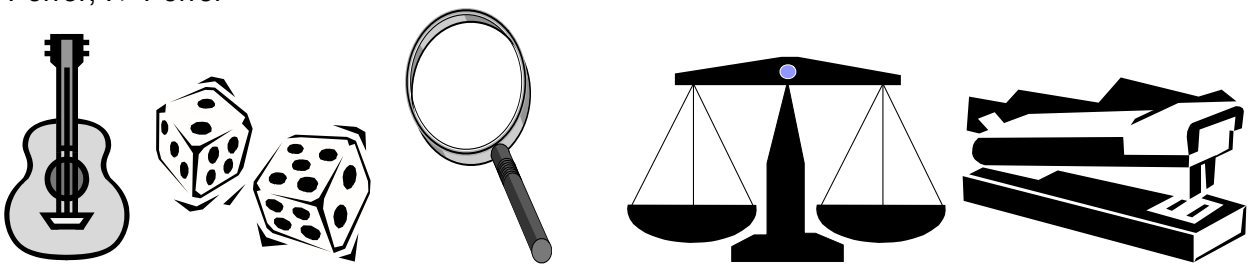
The Midland Cognitive Assessment (MICA) is a brief assessment tool that includes tasks for all of the 6 elements of cognition. It is still a work in progress and any feedback to the authors would be most welcome. The test is available on the following pages.

MICA (Midland Cognitive Assessment) Patient:_____ Date:_____

Handedness: _____ Assessment completed by:_____ (Version 17/5/16)

The MICA is a collection of cognitive tasks that can be done in full or in part. Emphasis is on cognitive domain assessment and not on total scores. For instructions see the accompanying booklet "Bedside Cognitive Assessment."

LANGUAGE COMPREHENSION Evaluate comprehension of spoken language by using multi-stage commands. **"Can you point with your finger to the ceiling, floor and the door."** N Normal) = 3 stages complete without an error; E (Equivocal) = some difficulty; I (Impaired) = 1 or more clear errors	**N E I**
WORKING MEMORY VERBAL TRIAL 1 **"Remember the following words."** Read a word and ask the patient to repeat it. Complete the set of 10 words. N > 6, E = 5-6, I < 5. CARROT BOAT LION EYE JUG CAT SOUP LEG HOUSE TICKET **"What were those words?"**	**N E I __/10**
SHORT-TERM MEMORY VERBAL TRIAL 2 Read the words again, asking for repetition after every word. **"What were those words?"** N > 6, E = 5-6, I < 5. CARROT BOAT LION EYE JUG CAT SOUP LEG HOUSE TICKET **"Now remember the words because I will ask you later for it."**	**N E I __/10**
VISUOSPATIAL & PRAXIS **"Copy these figures."** Do not warn about delayed recall. Score each drawing: 3 = no mistakes, 2 = few omissions, 1 = poor, 0 = no drawing. N > 6, E = 6, I < 6	
☜_____**Fold here so that the patient cannot see the above**_____☞	
 	**N E I ___/9**
ATTENTION **"I am going to read you a long series of letters. Whenever you hear the letter A, tap your hand like this."** Indicate tapping movement. N = no mistakes, E = one mistake and I = > 1 mistake. A mistake is an omission or inclusion. **A L T A A D A R S N A G A A A K P A T C A A P A L R**	**N E I**
EXECUTIVE ANIMAL NAMING TASK: **"I am going to give you one minute to name as many animals as possible."** The patient can name mammals, birds, reptiles or fish. Repetitions are not scored. Scoring: >14 = N; 12-14 = E; <12 = I	**N E I**
EXECUTIVE LURIA ALTERNATING HAND MOVEMENTS: **"Place both hands on the table like I do. Watch me carefully because I want you to do the same as I am doing. Can you see that I am making a fist with one hand? Now I swap it around. Can you do that as fast as you can?"** N = 3 cycles without any mistakes; E = 1-2 cycles; and I for not being able to complete the task.	**N E I**
EXECUTIVE SERIAL ORDER REVERSAL TASK: **"Say the months of the year starting with January."** Allow the person to complete the task until December. **"Now start with December and go backwards."** Stop the person when he reaches July or makes 2 or more mistakes: December November October September August July. N = no errors; E = 1 error; I > 1 error	**N E I**
SHORT-TERM MEMORY VERBAL DATE [ ] MONTH [ ] DAY [ ] PLACE [ ] CITY [ ] Date can be +/- one day, the rest must be exact. N = 5, E = 4, I < 4	**N E I ___/5**
PRAXIS **"Take this pen and rotate it with your fingers as fast as you can."** Test both hands. N = no errors; E = some difficulty; I = clear difficulty	**N E I**
SHORT-TERM MEMORY VERBAL **"What were the words I asked you to remember?"** N >5, E	**N**

= 5, I < 5 CARROT BOAT LION EYE JUG CAT SOUP LEG HOUSE TICKET	**E** **I** **__/10**
SHORT-TERM MEMORY VERBAL **"I am now going to read a list of words. Some of the words are from the list that I asked you to remember and some of the words are new. Please tell me which words were from the original list. If you don't know the answer just guess."** The instructions may be repeated if it is forgotten. **CARROT** ONION BUG **JUG** SHIP **BOAT LION** GOAT SPY **EYE SOUP** BUTTER BAT ARM **CAT LEG** MOUSE **TICKET HOUSE** BUCKET Give a mark for each word correctly identified (in bold) or rejected (words not in bold). N > 16, E = 14-16, I < 14	**N** **E** **I** **__/20**
SHORT-TERM MEMORY VISUAL **"Earlier, you copied three pictures, can you remember it? Can you draw it here?"** Provide the patient with a blank sheet of paper. 3 = no mistakes, 2 = few omissions, 1 = poor recall, 0 = a failure to recall anything. N > 5, E = 5, I < 5	**N** **E** **I** **___/9**
ANOMIA & AGNOSIA **"Name the following pictures."** If unable, ask the patient to describe what the object is used for. This is to distinguish anomia from agnosia. N = all correct; E = 1 error, I >1 error 	**Anomia N E I** **Gnosis N E I**
EXECUTIVE **"You have one minute to draw anything using four lines. The drawings must be different and I must be able to count the lines. I will give you two examples."** Draw a hash sign (#) and steps ⅂ . Count the lines as you draw them. **"Do you have any questions? Your minute starts now."** N > 7 drawings; E = 5-7 drawings; I < 5 drawings	
	N **E** **I**
SPOKEN LANGUAGE Estimate spoken language ability by evaluating speech. It should be clear, fluent and understandable. N = normal speech, E = equivocal, I definite impairment	**N** **E** **I**
MEMORY SUMMARY: WM VERBAL = ____/10; STM VERBAL = ____/45; STM VISUAL = ___/9; TOTAL MEMORY = ____/64	

COGNITIVE DOMAIN ASSESSMENT SUMMARY:		
LANGUAGE: SPOKEN = N E I, COMPREHENSION = N E I, NAMING = N E I	**ATTENTION = N E I**	**MEMORY = N E I**
EXECUTIVE = N E I	**PRAXIS = N E I** **GNOSIS = N E I** (Praxis and gnosis are not comprehensively covered in the MICA)	

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